



ArduSat



Astro Pi



ArduLab

Rockets: Possible design path

1. Design a basic rocket to launch from the ground (alpha)
2. Make it more powerful (beta)
3. Remove the fins and control via thrusting (candle)
4. (Dreamer?)

—Courtesy: openautics.com

features everything from the basics to features that enable you to understand the numerous sensors that can be placed on satellites and how to use these. It also makes individual space kits available, which include Arduino Uno, lessons on remote sensing, a collection of sensors and other components.

Astro Pi

Would you rather explore the universe on the earthy-named Raspberry Pi? The Raspberry Pi Foundation recently announced that they have teamed up with British European Space Agency (ESA) astronaut Tim Peake to send Raspberry Pis

to space, where these will be connected to the Astro Pi board. How cool is that!

Once in the ISS, these Raspberry Pi computers will be connected to Astro Pi boards, which come loaded with a collection of gadgets and, of course, sensors.

Astro Pi systems will be deployed in a number of different locations on board the ISS, where these will collect the generated data and download it to Earth, where these will be distributed to the teams.

Astro Pi allows students the chance to devise and code their own experiments or apps to run in space. The Astro Pi board or hardware attached on top (HAT) comes with a gyroscope, magnetometer sensor and an accelerometer, as well as a suite of sensors that can detect temperature, humidity and barometric pressure. There is also a camera module and

“I would like to die on Mars, just not on impact” — Elon Musk, CEO of SpaceX, at SXSW 2013

an infra-red camera module attached to it.

The foundation also set up a competition to promote code development, which is set under five themes, namely, spacecraft sensors, satellite imaging, space measurements, data fusion and space radiation. This event has several space agencies involved, such as the UK Space Agency, European Space Agency, ESERO-UK, Airbus DS, National Nuclear Laboratory and the National Physical Laboratory.

ArduLab

Created by Manu Sharma, co-founder of Infinity Aerospace, ArduLab is an ATmega2560-powered platform that enables NASA-approved experiments to be run inside the ISS. The result is a kit that is to space experiments what Arduino is to electronics experiments. It comes with all necessary features and interfaces to kick-start experimenting on the ISS. Atmel claims, on their website, that ArduLab has been used by both school students and NASA-JPL researchers for experimenting.

While it used to retail at more than ₹ 120,000 per kit, the good people at Infinity Aerospace later open sourced the entire project so that you can build it yourself. Their website ardulab.com has detailed plans that will help you to build the three main components of the system; the chassis, electronics and software (ArduLab OS). It also includes a kit that comes with some basic programs in Arduino Sketch File format (*.ino). Ino is a command line toolkit for working with Arduino hardware.

Now that you have all your space experiments set and ready, all you need to do is build a rocket to see how high you can get it.